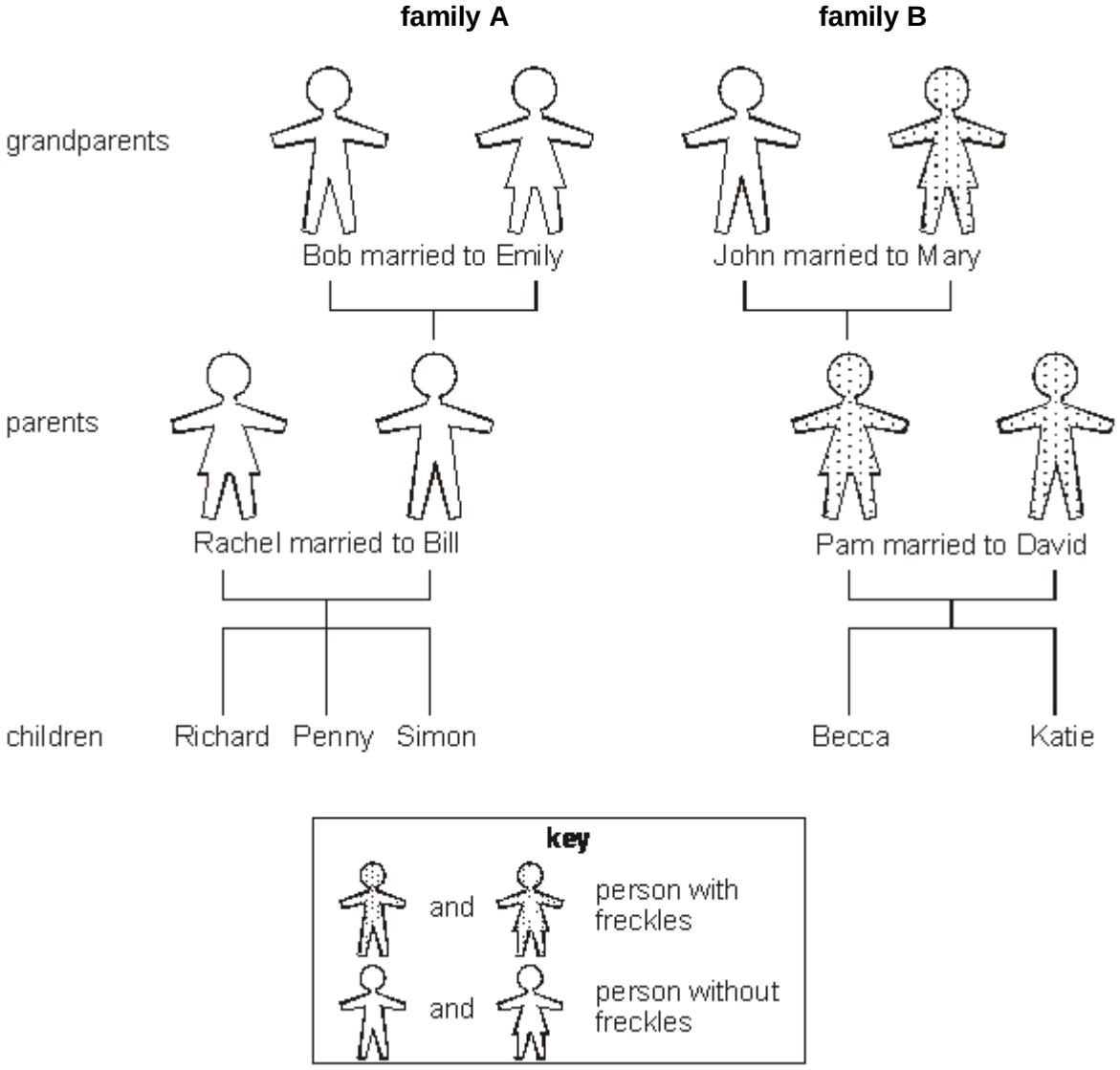
Q1.

The diagram shows two families. Some of the people in the diagram have freckles.



(a) (i) Which children are most likely to have freckles?
Tick the correct boxes.

Richard	Simon	Katie	Penny	Becca
☐	☐	☐	☐	☐

1 mark

(ii) How did you decide?

.....

.....

1 mark

(iii) Suggest why Bill does **not** have freckles.

.....

.....

1 mark

(b) (i) Which **two** cells pass on information from parents to their children?
Tick the **two** correct boxes.

bone cell	☐	cheek cell	☐
egg cell	☐	muscle cell	☐
red blood cell	☐	sperm cell	☐

1 mark

(ii) Which organ system produces these two cells?
Tick the correct box.

circulatory system	☐
digestive system	☐
reproductive system	☐
respiratory system	☐

1 mark
maximum 5 marks

Q1.

(a) (i) Katie ✓
Becca ✓

both *answers are required for the mark*

if more than two boxes are ticked, award no mark

1 (L3)

(ii) any **one** from

- their mother **or** Pam has freckles
- their father **or** David has freckles
- their parents have freckles
- their grandmother **or** Mary has freckles
- only family B has freckles

accept 'Rachel and Bill do not have freckles'
accept 'a grandparent has freckles'
*accept 'their family **or** the family on the right has freckles'*
accept 'freckles run in the family'
accept 'family A does not have freckles'

1 (L4)

(iii) his parents **or** Bob and Emily do not have freckles

accept 'his family does not have freckles'
accept 'his grandparents do not have freckles'
*accept 'he is not in family B' if the answer for **(ii)** is 'only family B has freckles'*

1 (L4)

(b) (i) egg cell ✓
sperm cell ✓

both *answers are required for the mark*

if more than two boxes are ticked, deduct one mark for each incorrect tick

minimum mark zero

1 (L4)

(ii) reproductive system ✓

if more than one box is ticked, award no mark

1 (L4)

[5]